Nonuniqueness for a reversible cubic reaction–diffusion system

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Abstract

We construct two global renormalized solutions with the same nonnegative finite-entropy initial data for the reversible cubic reaction $A + 2B \rightleftharpoons 3B$ on a three-dimensional ball. The two diffusion coefficients are equal. One species starts from zero, while the other has an integrable $|x|^{-1}$ singularity. The zero-species solution coexists with a second solution obtained from seeds whose initial masses tend to zero. A compact self-similar subsolution keeps the second solution nonzero. We prove its exact initial trace and the square-root Sobolev bounds needed for the full renormalized formulation. Thus entropy dissipation and local Lipschitz continuity of the reaction do not imply uniqueness for all nonnegative finite-entropy initial data.

1 Introduction and main result

Reaction–diffusion systems describe concentrations transported by diffusion and changed by local reactions. Their entropy inequalities often remain available when the concentrations are too irregular to make the reaction terms integrable. Renormalized solutions express the equations after smooth cutoffs in concentration space. For entropy-dissipating reactions, Fischer proved that these solutions coincide with a sufficiently regular comparison solution whose concentrations are bounded above and bounded away from zero uniformly [1, Definition 2 and Theorem 7, preprint version]. The distinction between this weak–strong statement and uniqueness of arbitrary renormalized solutions is also discussed in [2, Remark 1.4].

Here we consider nonnegative initial concentrations of finite entropy, allowing an unbounded concentration and an initially absent species. In this class, a reversible cubic reaction already gives nonuniqueness. Let $\Omega = B_1(0) \subset \mathbb{R}^3$ and consider

$$\begin{cases} a_t = \Delta a - b^2(a-b), \\ b_t = \Delta b + b^2(a-b), \end{cases} \qquad \partial_\nu a = \partial_\nu b = 0 \quad \text{on } \partial\Omega. \tag{1}$$

The reaction $A + 2B \rightleftharpoons 3B$, with unit forward and reverse rates, has reaction vector

$$f(a,b) = \big(-b^2(a-b),\ b^2(a-b)\big). \tag{2}$$

It preserves the nonnegative quadrant and satisfies, for every $a, b > 0$,

$$f_1(a,b)\log a + f_2(a,b)\log b = -b^2(a-b)(\log a - \log b) \le 0. \tag{3}$$

For entropy variables $\log u_i + \mu_i$, where the μ_i are fixed real constants, this gives the entropy sign with $\mu_1 = \mu_2 = 0$.

We use the following precise solution class. As usual, $0 \log 0 = 0$.

Definition 1.1. A nonnegative vector $u = (u_1, u_2) = (a, b)$ is a global renormalized solution of (1) with initial data u_0 if, for every $T < \infty$,

$$u_i \in L^\infty(0, T; L^1(\Omega)), \qquad \sqrt{u_i} \in L^2(0, T; H^1(\Omega)), \quad i = 1, 2, \tag{4}$$

and the following identity holds for every smooth $\xi : [0, \infty)^2 \to \mathbb{R}$ with compactly supported derivative $D\xi$, every smooth ψ on $\overline{\Omega} \times [0, T]$, and almost every terminal time T:

$$\begin{aligned} &\int_\Omega \xi(u(T))\psi(T)\,dx - \int_\Omega \xi(u_0)\psi(0)\,dx - \int_0^T \int_\Omega \xi(u)\partial_t\psi\,dx\,dt \\ &\quad = -\sum_{i,j=1}^{2} \int_0^T \int_\Omega \psi\xi_{ij}(u)\nabla u_i \cdot \nabla u_j\,dx\,dt - \sum_{i=1}^{2} \int_0^T \int_\Omega \xi_i(u)\nabla u_i \cdot \nabla\psi\,dx\,dt \\ &\qquad + \sum_{i=1}^{2} \int_0^T \int_\Omega \xi_i(u) f_i(u)\psi\,dx\,dt. \end{aligned} \tag{5}$$

Here $\nabla u_i = 2\sqrt{u_i}\nabla\sqrt{u_i}$ in the gradient products.

Theorem 1.2. *Set $C = 10000$ and*

$$a_0(x) = \frac{C}{|x|} \quad (x \neq 0), \qquad b_0(x) = 0, \tag{6}$$

with any finite nonnegative value assigned to $a_0(0)$. These data satisfy

$$\sum_{i=1}^{2} \int_\Omega u_{i,0}(1 + |\log u_{i,0}|)\,dx < \infty.$$

System (1) has at least two different global renormalized solutions with these initial data. If m is the Neumann heat evolution of a_0, one solution is $(m, 0)$. Another is $(m - b, b)$, where

$$0 \le b \le \frac{m}{2} \quad (t > 0), \qquad b(t, x) \ge t^{-1/2}\left(1 - \frac{|x|^2}{t}\right)_+^2 \quad \left(0 < t \le \frac{1}{4}\right). \tag{7}$$

Both solutions have the strong L^1 initial trace $(a_0, 0)$ and are regular on every compact positive-time interval.

The polynomial vector field is locally Lipschitz, and its boundary signs are $f_1(0, b) = b^3 \ge 0$ and $f_2(a, 0) = 0$. Thus the theorem gives an instance of an entropy-dissipating system with positive constant diffusivities in the usual finite-entropy comparison class. The singular initial concentration in (6) is part of the construction. Theorem 1.2 gives a negative answer to Problem 482 in the AIM collection [5], for its stated class of nonnegative finite-entropy data.

Equal diffusion is useful here: the total concentration $m = a + b$ solves a linear heat equation. It therefore bounds the nonlinear evolution globally, while its initial singularity permits a nonzero catalyst concentration to grow from a sequence of seeds with vanishing mass. The main point at the initial time is to show that this limiting concentration still has trace zero. A localized estimate away from the singular point proves that trace; the entropy estimate then gives (4).

2 A heat profile and a compact subsolution

Let $S(t)$ denote the Neumann heat semigroup on Ω, and put $m(t) = S(t)a_0$. Since $a_0 \in L^2(\Omega)$, positivity, strong continuity and positive-time regularity of the heat semigroup apply [3, Example 5.2]. In particular m is bounded on each $[\delta, T] \times \overline{\Omega}$, $\delta > 0$, and

$$m(t) \longrightarrow a_0 \text{ in } L^1(\Omega), \qquad \int_\Omega m(t)\,dx = 2\pi C. \tag{8}$$

The following explicit bounds control the behavior near the initial time.

Lemma 2.1. *For $t > 0$ and $x \neq 0$,*

$$\frac{C}{\sqrt{6t + |x|^2}} \le m(t,x) \le \frac{C}{|x|} + \frac{C}{2}|x|^2 + 3Ct. \tag{9}$$

Consequently, for each finite T,

$$\lim_{\rho \downarrow 0} \sup_{0<t\le T} \int_{B_{2\rho}} m(t,x)\,dx = 0. \tag{10}$$

Proof. For $\eta > 0$ set $m_{0,\eta} = C(|x|^2 + \eta)^{-1/2}$ and $m_\eta = S(t)m_{0,\eta}$. The function $\ell_\eta = C(6t + |x|^2 + \eta)^{-1/2}$ satisfies

$$(\partial_t - \Delta)\ell_\eta = -3C|x|^2(6t + |x|^2 + \eta)^{-5/2} \le 0, \qquad \partial_\nu \ell_\eta \le 0.$$

Its initial value is $m_{0,\eta}$, so $\ell_\eta \le m_\eta$ by comparison. For the upper bound, define

$$w_\eta = C(|x|^2 + \eta)^{-1/2} + \frac{C|x|^2}{2(1+\eta)^{3/2}}.$$

Then $\partial_\nu w_\eta = 0$ on $\partial\Omega$, $w_\eta \ge m_{0,\eta}$, and

$$\Delta w_\eta = -3C\eta(|x|^2 + \eta)^{-5/2} + 3C(1+\eta)^{-3/2} \le 3C.$$

Hence $m_\eta \le w_\eta + 3Ct$. These comparisons follow by testing the positive part of the heat-equation difference; the boundary term for $\ell_\eta - m_\eta$ is nonpositive. They only use strong L^2 initial traces of the regularized data. Letting $\eta \downarrow 0$ proves (9), since $m_{0,\eta} \to a_0$ in L^2. Its upper bound has integral $O(\rho^2) + O_T(\rho^3)$ on $B_{2\rho}$, which proves (10). □

The lower bound in (9) has the same parabolic scale as the following compactly supported function.

Lemma 2.2. *For $0 < t \le T_0 := 1/4$, let*

$$\underline{b}(t,x) = t^{-1/2}\Phi(x/\sqrt{t}), \qquad \Phi(y) = (1 - |y|^2)_+^2.$$

Then $0 \le \underline{b} \le m/2$, its support is contained in $\overline{B_{1/2}}$, and it is a weak subsolution of

$$b_t = \Delta b + b^2(m - 2b) \tag{11}$$

with zero Neumann boundary flux.

Proof. Write $s = |y|^2$ and $\mathcal{L} = \Delta_y + \frac{1}{2} y \cdot \nabla_y + \frac{1}{2}$. Direct calculation on $s < 1$ gives

$$\mathcal{L}\Phi = -\frac{23}{2} + 17s + \frac{5}{2}s^2, \qquad (\partial_t - \Delta)\underline{b} = -t^{-3/2}\mathcal{L}\Phi.$$

Also $m(t,x) \geq t^{-1/2} C/\sqrt{6+s}$ on the support of $\underline{b}$. Thus the desired subsolution inequality follows from

$$-\mathcal{L}\Phi \leq \Phi^2 \left(\frac{C}{\sqrt{6+s}} - 2\Phi \right). \tag{12}$$

For $s \leq 3/4$, the left side is at most $23/2$ and the right side is at least $(C/3 - 2)/256 > 23/2$. For $3/4 \leq s < 1$, one has $\mathcal{L}\Phi \geq 85/32 > 0$, while the right side is nonnegative. Moreover $2\Phi \leq 2 < C/\sqrt{7}$, proving $\underline{b} \leq m/2$. At $s = 1$ both Φ and its first derivatives vanish. The extension by zero therefore has no surface measure in its distributional Laplacian, and (12) proves the weak inequality across the moving front. The support radius is $\sqrt{t} \leq 1/2$, so the outer boundary flux is zero. □

3 A second solution and its initial trace

For $0 < \varepsilon < T_0$, start (1) at time ε with

$$(a_\varepsilon, b_\varepsilon)(\varepsilon) = (m(\varepsilon) - \underline{b}(\varepsilon), \underline{b}(\varepsilon)). \tag{13}$$

These data belong to $W^{2,p}(\Omega)^2$ with zero Neumann trace for every finite p. Indeed, the compact seed is C^1, has bounded piecewise second derivatives, and vanishes near $\partial\Omega$. Bounded-data semilinear Neumann theory gives a maximal regular solution [4, Example 3.7 and Theorem 3.9]. Its total is m, by uniqueness of the scalar heat equation. Thus b_ε solves (11).

On each interval bounded away from zero the scalar reaction is locally Lipschitz in b. Both 0 and $m/2$ solve (11); comparison and Lemma 2.2 yield

$$0 \leq b_\varepsilon(t) \leq m(t)/2 \quad (t \geq \varepsilon), \qquad b_\varepsilon(t) \geq \underline{b}(t) \quad (\varepsilon \leq t \leq T_0). \tag{14}$$

For the subsolution comparison, testing the positive part of the difference gives its squared L^2 norm a differential inequality with a finite local Lipschitz constant. No bound on this constant as $\varepsilon \downarrow 0$ is needed. Since $a_\varepsilon = m - b_\varepsilon$, the bound by m excludes finite-time blowup of either component. Each seeded solution is global.

We spell out the positive-time compactness used to remove the seed time. Fix $0 < \delta < T$ and $\varepsilon < \delta/2$. The states and their polynomial reaction terms are uniformly bounded on $[\delta/2, T]$. Put $X = L^6(\Omega)^2$, let $A = \operatorname{diag}(\Delta_N, \Delta_N)$, and write $X_\theta = D_A(\theta, 6)$ for its real interpolation spaces. The mild formula and analytic smoothing imply

$$\begin{aligned} \|(a_\varepsilon, b_\varepsilon)(\delta)\|_{X_{2/3}} &\leq K_\delta \, \|(a_\varepsilon, b_\varepsilon)(\delta/2)\|_X \\ &\quad + K_\delta \int_{\delta/2}^{\delta} (\delta - s)^{-2/3} \, \|f(a_\varepsilon(s), b_\varepsilon(s))\|_X \, ds. \end{aligned}$$

The right side is uniform in ε. The embedding $X_{2/3} \hookrightarrow X_{1/2}$ is compact and $X_{1/2} \hookrightarrow C(\overline{\Omega})^2$ is continuous; see [4, Proposition 2.13, Theorem 2.14 and Example 3.7]. Extract a convergent subsequence at δ, then use continuous dependence for the bounded positive-time problem. Its limit remains bounded by m, so it extends to all later times. Nested subsequences at times decreasing to zero agree on overlaps and give one global regular solution

$$(a, b) = (m - b, b), \qquad 0 \leq b \leq m/2, \qquad b \geq \underline{b} \quad (0 < t \leq T_0). \tag{15}$$

The convergence at each positive time is uniform on $\overline{\Omega}$.

Lemma 3.1. *The solution in (15) satisfies $b(t) \to 0$ and $a(t) \to a_0$ in $L^1(\Omega)$ as $t \downarrow 0$.*

Proof. Fix $0 < \rho < 1/4$ and a smooth $0 \le \chi_\rho \le 1$ which is zero on B_ρ and one outside $B_{2\rho}$. It is constant near $\partial\Omega$. For each finite T, Lemma 2.1 bounds m by some $M_{\rho,T}$ on $0 < t \le T$, $|x| \ge \rho$. Set $Y_\varepsilon(t) = \int_\Omega b_\varepsilon(t)\chi_\rho\,dx$. Since $b_\varepsilon^2(m - 2b_\varepsilon) \le m^2 b_\varepsilon$,

$$Y_\varepsilon'(t) \le K_\rho + M_{\rho,T}^2 Y_\varepsilon(t), \qquad K_\rho = \tfrac{1}{2}\,\|\Delta\chi_\rho\|_\infty \int_\Omega a_0\,dx.$$

The diffusion term is $\int b_\varepsilon \Delta\chi_\rho$; both boundary terms vanish. If $\varepsilon < \rho^2$, the seed is supported in B_ρ, so $Y_\varepsilon(\varepsilon) = 0$. Hence $Y_\varepsilon(t) \le K_\rho t \exp(M_{\rho,T}^2 t)$. Pass to the limit at fixed $t > 0$ to obtain

$$\int_\Omega b(t)\,dx \le \frac{1}{2}\int_{B_{2\rho}} m(t)\,dx + K_\rho t \exp(M_{\rho,T}^2 t).$$

First let $t \downarrow 0$ with ρ fixed, and then let $\rho \downarrow 0$ using (10). This proves the zero trace of b. The trace of $a = m - b$ follows from (8). □

4 Entropy and the renormalized identity

Let $h(s) = s\log s - s + 1$ for $s > 0$, with $h(0) = 1$, and put $H(t) = \int_\Omega [h(a(t)) + h(b(t))]\,dx$. Since $a + b = m$, one has

$$0 \le h(a) + h(b) \le h(m) + 1. \tag{16}$$

For any fixed finite T, (9) dominates the right side on $0 < t \le T$ by a constant multiple of $1 + |x|^{-1}\log(2 + |x|^{-1})$, an integrable function on the ball. Lemma 3.1, continuity of h and dominated convergence therefore give

$$H(t) \longrightarrow H_0 := \int_\Omega [h(a_0) + h(0)]\,dx < \infty. \tag{17}$$

For clarity, apply dominated convergence along an almost-everywhere subsequence of any sequence converging to zero in time. Every such sequence has this subsequence by the strong L^1 trace, so the full limit follows.

For $\eta > 0$ use the smooth entropy $h_\eta(s) = h(s + \eta)$ on $[\delta, T]$, where $\delta > 0$. Write $H_\eta(t) = \sum_i \int h(u_i(t) + \eta)$. The equations and the Neumann condition give

$$\begin{aligned} H_\eta(T) &+ \sum_{i=1}^{2}\int_\delta^T\int_\Omega \frac{|\nabla u_i|^2}{u_i + \eta}\,dx\,dt \\ &+ \int_\delta^T\int_\Omega b^2(a - b)\big[\log(a + \eta) - \log(b + \eta)\big]\,dx\,dt = H_\eta(\delta). \end{aligned} \tag{18}$$

The last integrand is nonnegative because $a \ge b$. This calculation is valid even where $b = 0$. It bounds $\sqrt{u_i + \eta}$ in $L^2(\delta, T; H^1(\Omega))$, while $|\sqrt{u_i + \eta} - \sqrt{u_i}| \le \sqrt{\eta}$ gives strong L^2 convergence. Weak compactness and lower semicontinuity, followed by $\eta \downarrow 0$, yield

$$4\sum_{i=1}^{2}\int_\delta^T\int_\Omega |\nabla\sqrt{u_i}|^2\,dx\,dt \le H(\delta) - H(T).$$

Now let $\delta \downarrow 0$ and use (17). Together with $a + b = m$ and its conserved mass, this proves (4) for every finite T. The gradient identity used in Definition 1.1 follows as well: test $\nabla u_i = 2\sqrt{u_i + \eta}\nabla\sqrt{u_i + \eta}$ on a fixed positive-time strip, and pass to the limit using strong L^2 convergence of the first factor and weak L^2 convergence of the second.

Proof of Theorem 1.2. The initial entropy in (6) is finite, since its radial integral is a constant multiple of $\int_0^1 r[1+\log(C/r)]\,dr$. The construction above gives a global solution $(m-b,b)$ with the required initial trace and square-root Sobolev bounds. It remains to check the full identity (5) at the initial time.

For fixed $\delta>0$, the smooth chain rule gives that identity on $[\delta,T]$. On the common compact support of $D\xi$ and $D^2\xi$, the components u_i are bounded by some R. The identity $\nabla u_i = 2\sqrt{u_i}\nabla\sqrt{u_i}$ then bounds every Hessian term by a constant times $|\nabla\sqrt{u_i}|\,|\nabla\sqrt{u_j}|$ and every flux term by a constant times $|\nabla\sqrt{u_i}|$. These terms are integrable on $(0,T)\times\Omega$. The cutoff reaction term is bounded. Also ξ is globally Lipschitz, so the strong L^1 initial trace gives $\xi(u(\delta))\to\xi(u_0)$ in L^1. Letting $\delta\downarrow 0$ therefore proves (5) for every permitted ξ and ψ, in fact for every $T>0$.

The second solution $(m,0)$ has zero reaction, the same strong initial trace, and the same entropy dominator (16). The regularized entropy calculation (18) applies to it with zero reaction term, giving its Fisher bound without testing $\log 0$. The identical chain-rule passage proves (5) for this solution as well. Finally $b\geq\underline{b}>0$ on $0<t<T_0$, $|x|<\sqrt{t}$, a set of positive space-time measure. The two renormalized solutions are different. □

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